

TOPICAL REVIEW

Challenges associated with the wire arc additive manufacturing (WAAM) of aluminum alloys

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Challenges associated with the wire arc additive manufacturing (WAAM) of aluminum alloys

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E-mail: shivraman.fmc@kiit.ac.in and thapliyalr@gmail.com**Keywords:** wire arc additive manufacturing, aluminum alloys, solidification cracking**Abstract**

The wire arc additive manufacturing (WAAM) has been widely used for the development of aluminum alloy based components and parts used in the aerospace and automobile industries. Nevertheless, the development being at a nascent stage, the comprehensive view on microstructure and mechanical characterization of WAAM built is the need of the hour. Hence, this review paper deals with the microstructural and mechanical characterization of WAAM built aluminum alloys. The review covers the effect of process parameters on the soundness of built and it further provides insight into the difficulties encountered during the process and the future perspective of the WAAM built aluminum alloys.

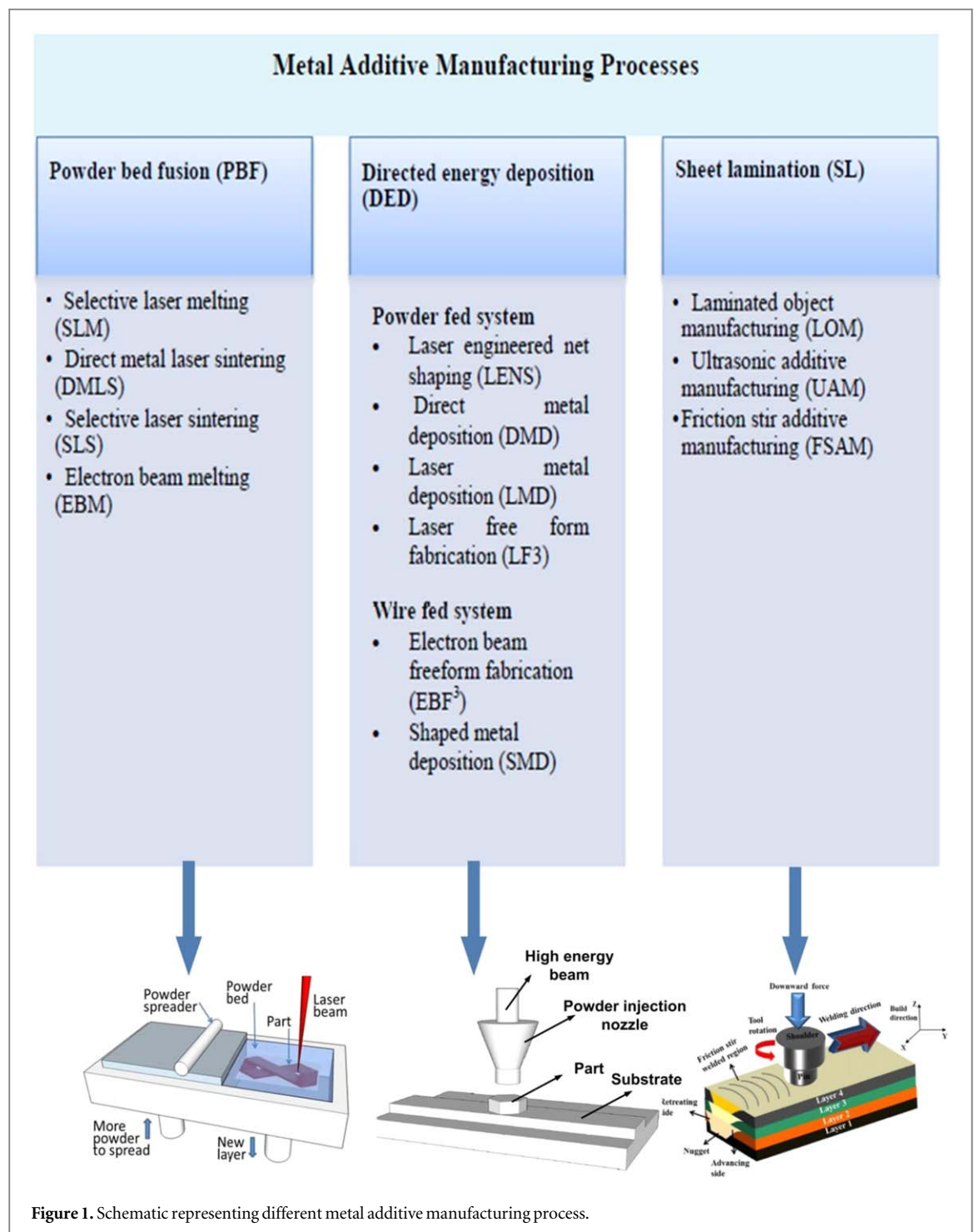
1. Introduction

The development of a component for better mechanical properties with lesser cost and delivery time has always been a major challenge for automobile and aerospace industries. Recently, the additive manufacturing process has been found to be a plausible method for dealing with complexity related to the manufacturing of different parts for these industries with less material wastage and delivery time. Therefore, the aerospace industries and research organization are keen to study this new manufacturing process for wider applications and adaptability [1–3].

Additive manufacturing (AM) is the process of developing a three dimensional (3D) products through the layer-by-layer deposition of the material using the aid of CAD/CAM model. In the early 1980s, AM was utilized for faster product development and was termed as rapid prototyping. The commercialization of this process began in the early 1990s owing to its ability to develop the metallic parts of intricate shapes in fewer steps and less lead-time successfully. The ability of AM to develop the complex metallic parts shifted the paradigm towards the utilization of this process for developing complex structure and parts. Till date, additive manufacturing has been successfully utilized for fabrication of various metals and polymers but this paper specifically covers additive manufacturing for metals only.

The metal AM utilizes laser beam, electron beam and arc as a heat source and material in the form of wire and powder form is used as the feedstock material. The categorization of the process involved in AM of metal is shown in figure 1. However, the anisotropy in the microstructure, mechanical properties, and defects in the metallic AM process exhibits extreme challenges to the acceptability of the process [4–6]. Therefore, the constant efforts for upgrading the various methodologies to develop the sound metallic AM process in terms of research and publications are in progress [7–9].

The details of the principle of the above-listed processes are very well addressed in the previously published reviews [8, 10–14]. This review deals with the scope of wire arc additive manufacturing (WAAM) process for the development of aluminum-based components for the automobile and aerospace industries and analyses the metallurgical and mechanical characterization of the WAAM built aluminum alloys. In addition, the effect of process parameters on the microstructural and mechanical properties of the WAAM built aluminum alloys and the defects encountered in the process have been comprehensively discussed.



2. Wire arc additive manufacturing process for aluminum alloys

The wire arc additive manufacturing (WAAM) comes under the category of direct energy deposition (DED) techniques in which wire is used as the feedstock material that provides higher mass flow and deposition rate than the powder-based feedstock material. The wire arc additive manufacturing (WAAM) process utilizes the working principle of the arc welding process, i.e., Gas tungsten arc (GTA), Gas metal arc (GMA), and Cold metal transfer (CMT). The material in wire forms is easier to store and manufacture through certified processing methods. The advantages of WAAM over other processes are low capital cost, high deposition rate, and ease in manufacturing the large parts. However, defects like porosity, distortion, and residual stresses are major challenges. Nevertheless, the design and development of components and parts for aerospace and automobile industries have aggravated with the rising research interest in this process.

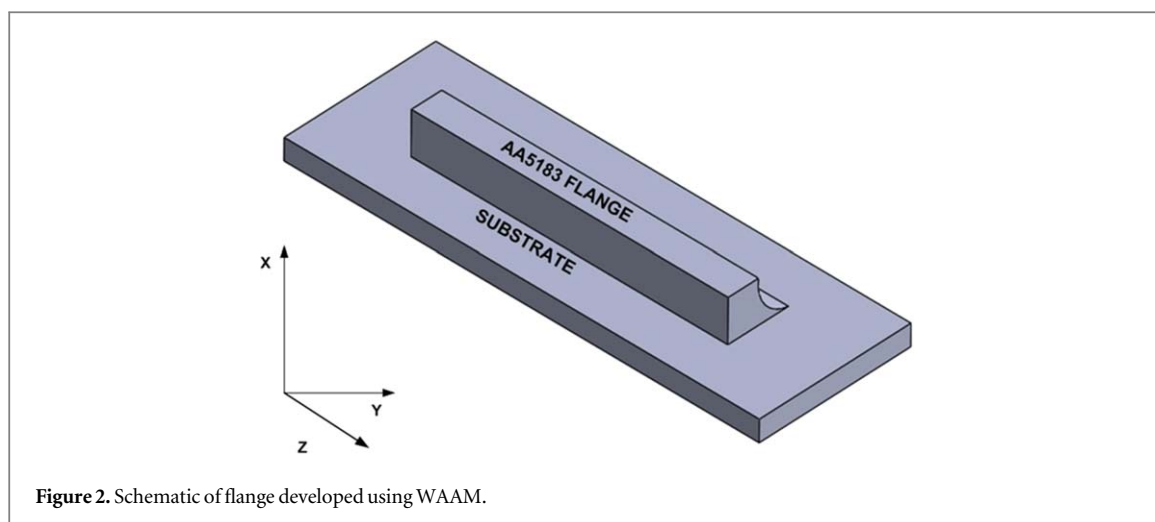


Figure 2. Schematic of flange developed using WAAM.

2.1. Metallurgical and mechanical characterization

Wang *et al* [15] developed the Al4043 alloy component using gas tungsten arc welding (GTAW) and established that the increase in the heat input results in expansion of the bead width and higher penetration (less height). The microstructure of the top region exhibited the equiaxed structure with a maximum hardness, while the bottom and central portion exhibited the coarser grain having an average hardness value of 42 HV. The grain boundaries and interdendritic regions were characterized by the presence of precipitates [16].

A flange type additive manufactured AA-5813 was developed using the short-pulsed mode GTAW in 70% He-30% Ar shielding atmosphere exhibited a mixed microstructure having equiaxed and columnar grains with the hardness of $70\text{--}75\text{ kg mm}^{-2}$ and $73\text{--}78\text{ kg mm}^{-2}$ along the X and Y directions (figure 2). The average mean strength of the developed parts was 293 MPa with a slightly lower value in the X-direction as compared to the Y-direction because mixed microstructure in X-direction [17].

Recently, the cold metal transfer (CMT) process have been used for WAAM process because this process offers low heat input and less spattering tendency resulting in better mechanical properties. Cong *et al* [18] developed multilayered AA2319 parts under different CMT modes. However, the authors ruled out the conventional CMT method for multilayer deposition owing to a large number of porosities encountered during single layer deposition. The AM part developed by pulse advance mode exhibited the lowest porosity owing to the lower heat input and effective oxide cleaning of welding wire. The low heat input decreases the hydrogen solubility in the molten pool during the process due to lower peak temperature. Additionally, the high cooling rate offered by the pulse advance mode results in fine grain structure, which reduces the grain boundary liquation. The wire arc manufactured Al-6Mg alloy under variable polarity (CMT mode) exhibited the maximum strength of 333 MPa, which is higher than the wrought alloy system because of the fine and equiaxed grain structure throughout the built [19].

The formation of pores due to hydrogen solubility and entrapment of gases is a serious issue in WAAM process especially for the aluminum alloys. Hence, efforts have been made to minimize the formation of pores in the aluminum alloys during WAAM [20]. The interlayer rolling during the WAAM process is used to reduce to the pore formations in the additive manufactured parts. The interlayer rolling causes the annihilation of large grain boundaries into small-angle grain boundaries, which accumulates the dislocation and exerts resistance to the dislocation movement Furthermore, the precipitate strengthening also, contributes in the strengthening of the AM parts. The interlayer rolling causes the breaking of coarser particles into smaller particles and leads to the homogenous distributions of the second phase particles, which resists the dislocation movement and strengthens the AM product.

The deformation induced by rolling during WAAM Al5087 alloy decreases the pores formation caused by the absorption of hydrogen present in grain interiors through the induced dislocations (figure 3). However, the subsequent heat treatment during WAAM results in the opening and growth of the pores owing to the thermal motion and coalescence of secondary pores formed during the solidification of Al5087 alloy [20].

The pore closing due to interlayer rolling enhances the mechanical properties of WAAM product as compared to the as-deposited WAAM products [21]. However, the heat treatment (T6) of as-deposited additive manufactured AA2219 has exhibited comparable mechanical properties with the interlayer rolled and heat-treated sample. The improvement in the mechanical properties owing to heat treatment is attributable to the homogenous microstructure and reduction in dendrites present within the grain. The heat treatment result in

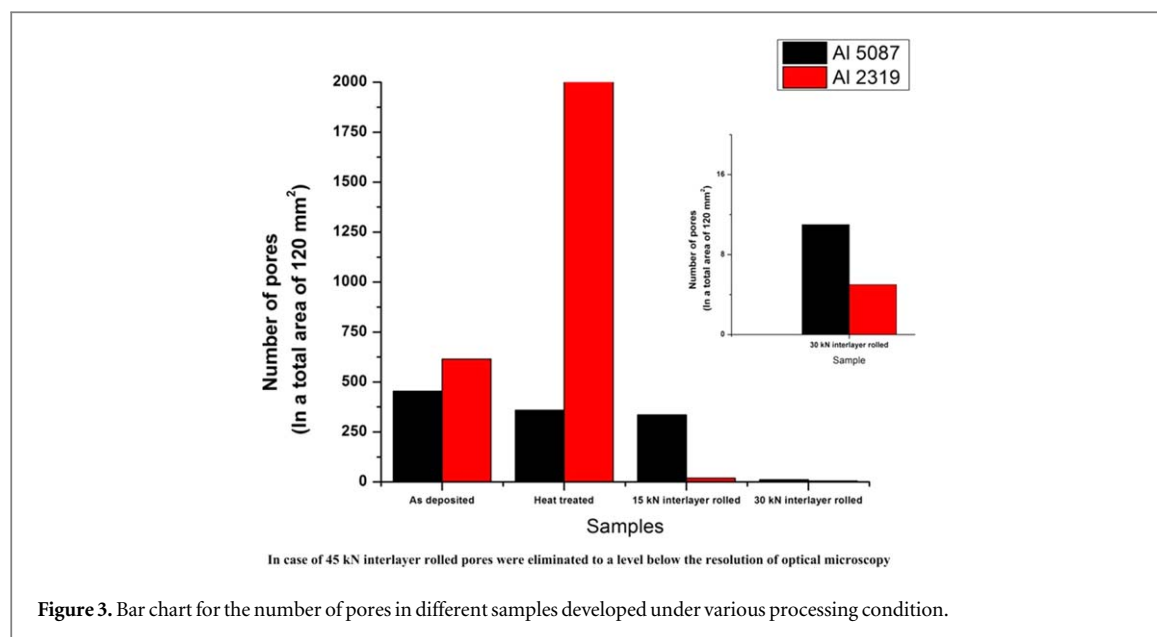


Figure 3. Bar chart for the number of pores in different samples developed under various processing condition.

fine a needle-like precipitate that provide higher resistance to the dislocation movements and consequently improves the mechanical properties.

The GTAW process for AM of the aluminum is found to be beneficial over GMAW and CMT process because the two different energies (i) TIG Torch and (ii) material source (wire feeding) which provides better control over the input parameters. However, the gap defect due to shifting of the weld bead from the original position is one of the major challenges encountered during GTAW process. Geng *et al* [22] reported that increasing the angle (Θ) between the wire and deposited surface shifts the landing of molten metal away from the arc axis, thereby resulting in the gap defect (figures 4(a)–(b)).

Increasing the distance between the feeding wire and substrate (H) (deposited layer), there is increase in the offset from the start position (gap defect) owing to the transformation of bridging the metal transfer mode into globular transfer mode (figure 4(c)). However, the decrease in the gap defect with increase 'H' (0.5 to 1 mm) is attributable to increase in the size of the molten metal droplet, which upon dropping on the surface leads to decrease the gap defect.

3. Defects and process parameter relation

The past work suggests that a close control over processing parameters is necessary to develop aluminum alloy without defects by WAAM. The defects encountered during the WAAM of aluminum alloys are similar to the defects encountered during the welding of aluminum alloys. This section covers the defects encountered during WAAM of aluminum and covers, the remedies to be adopted for the development of sound AM built with better mechanical properties.

3.1. Inhomogeneous microstructure

The complex thermal cycle encountered during the WAAM processing results in mix microstructure (coarse and fine grain) throughout the built, which is detrimental for the mechanical properties. The microstructure of aluminum AM built varies from columnar at the bottom to coarse equiaxed at the top depending upon the heating and cooling cycle encountered [11, 12, 15, 16]. The microstructure developed during AM can be explained using constitutional supercooling theory [23]. The microstructure developed during welding depends on the ratio of the temperature gradient (G) and growth rate (R) (figures 5(a)–(b)). The higher degree of constitutional supercooling, i.e., high G/R ratio results in greater degree of grain refinement [24].

A homogenous microstructure during WAAM can be obtained by varying the process parameters and process selections [19]. An equiaxed refine structure was obtained using arc pulsation owing to heterogeneous and surface nucleation mechanism which increases the grain nucleation sites [25]. The high cooling rate, caused by undercooling during the low current cycle of pulsed arc promotes these two mechanisms [25–27].

The selection of welding process also governs the grain refinement in WAAM because during melting, the weld pool encounters four driving forces i.e. surface tension force (F_T), electromagnetic force (F_E), buoyancy force (F_B) and arc shear force (F_S) which governs weld pool geometry and grain structure (figure 6). The effect of this process varies with the selection of welding process i.e. GTAW, GMAW and CMT.

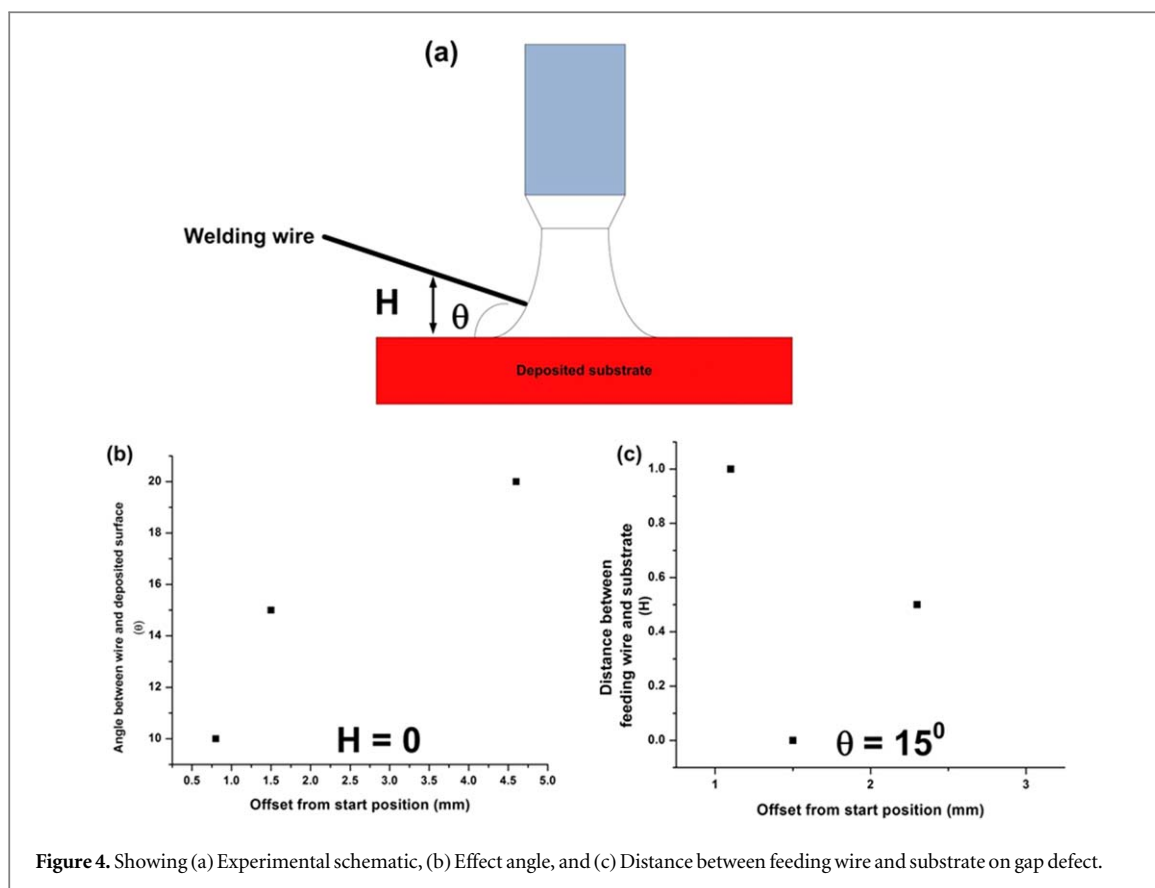


Figure 4. Showing (a) Experimental schematic, (b) Effect angle, and (c) Distance between feeding wire and substrate on gap defect.

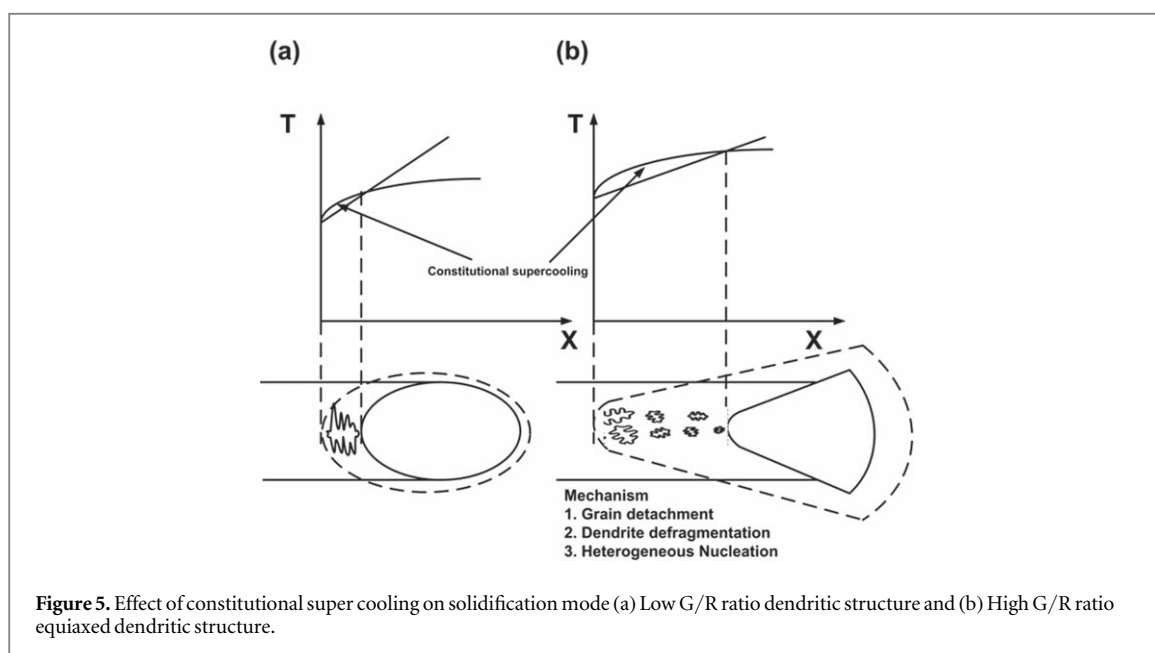
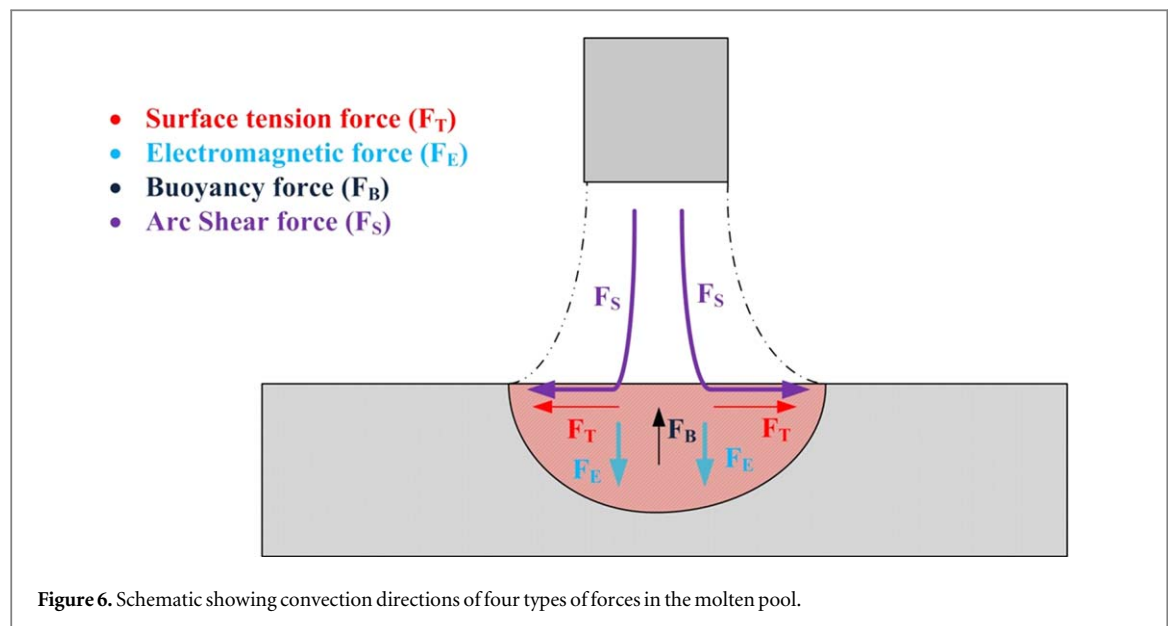


Figure 5. Effect of constitutional super cooling on solidification mode (a) Low G/R ratio dendritic structure and (b) High G/R ratio equiaxed dendritic structure.

In wire arc additive manufacturing of aluminum alloys effect of buoyancy force (F_B) is neglected due to shallower depth of the welding pool irrespective of the process selection [28]. Additionally, arc shear force (F_S) is ignored because the welding current is below 300 A in most of literature work published till date [29]. Therefore, the grain structure of the aluminum products developed by WAAM will depends on surface tension (F_T) and electromagnetic force (F_E). The effect of surface tension force on grain size (F_T) for three processes i.e. GTAW, GMAW, and CMT was moreover the same. Hence, the grain refinement is solely dependent on the electromagnetic force (F_E) which found to be higher in case of CMT as compared to GMAW and GTAW process [30]. The electromagnetic force provide stirring effect in the weld pool resulting in grain detachment and dendrite defragmentation, which in turn increases nucleation site and resists the formation of coarse columnar



structure [24, 31, 32]. Therefore, the combination weld pool stirring and undercooling effect in the pulsed CMT mode is a prominent reason for the homogeneous microstructure in the WAAM processing.

3.2. Porosity

This is the most common problem associated with WAAM due to dissolution and entrapment of gases during welding. The porosity can range from microporosity to coarse pores. The main agent for porosity in the aluminum is hydrogen and it has been found that the solubility decreases rapidly during the terminal stage of solidification [33]. The porosity formation depends on process parameters, alloy composition, and welding process which is discussed as follow as.

3.2.1. Effect of process parameters

The process parameters of WAAM include current, voltage, travel speed and surface condition of feedstock. The current voltage and travel speed governs the heat input. It has been established that higher heat input during the WAAM results in coarser microstructure with lesser grain boundaries which promotes liquation along the grain boundaries and gas entrapment [18, 25]. Also, an increase in the welding speed during the rapid prototyping of AA 5356 causes porosity in the AM built owing to the high solidification rate [16]. Additionally, an oxidized feed wire used for WAAM exhibited higher porosities due to presence higher concentration of oxygen in the pool [34].

3.2.2. Role of alloying elements

The alloying elements affects the hydrogen solubility in liquid aluminum [35]. The AM built of AA5087 (Mg-rich) exhibits a lesser tendency of pore formation than AA2319 (Cu-rich) alloy [20]. The pore formation in Al 5087 was found to be lesser than Al 2319 alloy because (i) the lower number for eutectic formation in the Mg-based alloy and (ii) the increase in the hydrogen absorption owing to the presence of copper in AA2319 that promotes formation of Al₂Cu precipitates. Additionally, the presence of copper decreases the hydrogen solubility in molten aluminum alloys which promotes porosity formation during solidification [36].

3.2.3. Process selection

The selection of welding process i.e. GTA, GMA and CMT dominant mechanical properties of the aluminum product developed by WAAM. Liu *et al* [37] reported that the GTAW welding of Al5083 results higher strength and hardness than the GMAW welds because of precipitate strengthening and lesser porosities formation. Similar observation were made during WAAM processing of aluminum alloys that the close control over material and heat input in the GTAW process than the GMAW process [22].

Further, the cold metal transfer (CMT) for WAAM of aluminum offers better mechanical properties by providing close control over the porosity formation, controlled metal deposition and low thermal heat input [38, 39]. The lower heat input during welding results in lower weld pool temperature and fine grain structure, which avoids the entrapment of hydrogen and thus the porosity formation [32, 40, 41].

3.3. Solidification cracking

The solidification cracking is a defect specifically observed during WAAM of the aluminum products. This occurs primarily due to the wide solidification temperature range of the aluminum alloys [42]. The other responsible reasons for solidification cracking are alloying elements, the amount of grain boundary liquid, and grain structure [41–43]. Ouyang *et al* [16] reported the solidification cracking for the 5356 AM products was due to the coarse grain structure and segregation of coarse second phase particles in the grain boundaries. The presence of low melting phases at the grain boundaries during the WAAM of AA5183 results in the intergranular type of hot cracking [17]. Kou *et al* [44] proposed that the fine equiaxed grain structure is less susceptible for solidification cracking because the fine grain structure being more ductile can easily accommodate the contraction strain and exhibit resistance to crack propagation. Furthermore, a fine grain structure has greater tendency of feeding and healing the incipient crack during solidification. Additionally, the degree of restraint governs the crack propagation during terminal stage of solidification; hence, the design and selection of AM fixture is a prerequisite.

3.4. Residual stress and distortion

Residual stresses are the internal stresses that remain inside the structure even after the removal of the load. The large thermal gradients during AM results in residual stress in the WAAM built. The residuals are detrimental for the mechanical properties when exceeds the local yield stress of material resulting in the plastic deformation [45]. Debroy *et al* [12] pointed out the key factors that are (1) thermal gradient, (2) thermal contraction and expansion due to cooling and heating, and (3) the coefficient of thermal expansion responsible for the residual stress during additive manufacturing. Distortion arises in the AM part because of the thermal cycles and improper fixing of the substrate.

The mechanism for residual stress development is similar for both fusion welding and the WAAM process [46, 47]. There are specific studies on the residual stress in titanium alloy developed by WAAM [48–51]. However, in case of aluminum AM products, the residual studies are constricted to part developed by SLM and EBM [52–54]. However, the residual stress during WAAM of aluminum alloys has not been well reported.

4. Concluding remarks and future scope

The automotive and aerospace industries are adapting the WAAM based AM techniques over the conventional machining and fabrication techniques for the development of complex geometries. The researchers have shown that the product developed using WAAM exhibited mechanical properties comparable or sometimes better than the wrought or cast products of aluminum. The microstructural and mechanical heterogeneity in WAAM processed aluminum alloys is still a major concern owing to the grain morphology, porosity, solidification cracking, and residual stress. However, the close control over parameters (current, voltage), process selection (arc pulsation and CMT mode), and process advances (interlayer rolling) have been utilized to address microstructural homogeneity and mechanical properties improvement. The lower heat input and faster cooling rates results in the equiaxed structure, with lesser porosity and better resistance to solidification cracking.

Currently, the WAAM of aluminum alloys is at a very nascent stage but the researches clearly suggest that this process is having promising future and with the development of newer technologies and hardware development, the process is catching pace with other conventional processes. The in-process modification of WAAM process like interlayer rolling, friction stir additive manufacturing (FSAM) and vibration assisted WAAM will find greater adoption and implementation of the process [14, 55, 56]. However, there is a critical need for understanding the mechanism of WAAM processed aluminum using mathematical tools. The investigation should further focus fabrication of the commercial alloys like 7XXX, used for aerospace and automobile industries. The wear and corrosion behavior of the WAAM built aluminum alloys need to be investigated because the aluminum alloys based components are highly used in area where it encounter wear and corrosion during its service life. Additionally, the efforts should be made to develop the product with faster deposition rates, lesser time, and quality control.

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